420-SN1-RE Final
John Abbott College
Fall 2024

Name: ____________________
Student ID: ____________________

No notes, no computers. Language dictionaries (hard-copy) allowed.
ANSWER ALL THE QUESTIONS ON THIS ANSWER SHEET.

1. (4 points) **Tracing** Output:

2. (4 points) **Powers of 2**

```
def main():
```

3. (8 points) **Pasta**

```
def main():
    pasta_lengths = [...]
```

4. (8 points) **Barnacles**

```
import statistics
def clean_data(barnacles: list) -> list:
    sigma = statistics.stdev(barnacles)
    average = sum(barnacles)/len(barnacles)
```

5. (3 points) **Inputs**

Give me an a:

Give me an b:

Give me an c:

6. (8 points) **Helium Balloons**

```
MAX_BALLOON_SIZE=15.0

def main():
    n = 42
    R = 8.31
    pressures = [...]   # all values > 0
    heights =   [...]
    temps =     [...]
```

7. (6 points) **Math Functions**

(assume that $b^2 > 4ac$)

```
def function1(a: float, b: float, c: float) -> float:
```

```
def function2(x: float) -> float:
```

```
def function3(v_i: float, v_f: float, t: float) -> float:
```

(don't worry about radians/degrees in this next function)

```
def function4(w: float, t: float, p: float) -> float:
```

8. (8 points) **Windows**

```
def main():
    wall_x1 = 0
    wall_y1 = 0
    wall_x2 = 12
    wall_y2 = 11
    big_x1 = 1
    big_y1 = 4
    big_x2 = 7
    big_y2 = 8
    small_x1 = 1
    small_y1 = 10
    small_x2 = 8
    small_y2 = 11
```

9. (8 points) **Cars**